

$$\begin{aligned}
12) \rightarrow & \frac{1}{16\pi^2} \left(-\frac{1}{9} g_2^4 \text{LF}_{3,0}[\mathbf{m}_2] - \frac{1}{6} g_2^4 \text{LF}_{4,-1}[\mathbf{m}_2] + \frac{8}{45} g_2^4 \text{LF}_{5,-2}[\mathbf{m}_2] + \right. \\
& \left(-\frac{1}{4} g_2^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \frac{5}{72} \sum_{\mathbf{p}} g_2^4 \right) \text{LF}_{3,0}[\mathbf{m}_l^{\text{p}}] + \left(\frac{1}{4} g_2^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} - \frac{1}{48} \sum_{\mathbf{p}} g_2^4 \right) \text{LF}_{4,-1}[\mathbf{m}_l^{\text{p}}] - \\
& \frac{2}{45} \sum_{\mathbf{p}} g_2^4 \text{LF}_{5,-2}[\mathbf{m}_l^{\text{p}}] + \left(-\frac{3}{4} g_2^2 \langle \overline{y_d^{\text{pr}}} y_d^{\text{pr}} + \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \rangle + \frac{5}{24} \sum_{\mathbf{p}} g_2^4 \right) \text{LF}_{3,0}[\mathbf{m}_q^{\text{p}}] + \\
& \left(\frac{3}{4} g_2^2 (\overline{y_d^{\text{pr}}} y_d^{\text{pr}} + \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) - \frac{1}{16} \sum_{\mathbf{p}} g_2^4 \right) \text{LF}_{4,-1}[\mathbf{m}_q^{\text{p}}] - \frac{2}{15} \sum_{\mathbf{p}} g_2^4 \text{LF}_{5,-2}[\mathbf{m}_q^{\text{p}}] - \\
& \frac{1}{18} g_2^4 \text{LF}_{3,0}[\tilde{\mu}] - \frac{1}{12} g_2^4 \text{LF}_{4,-1}[\tilde{\mu}] + \frac{4}{45} g_2^4 \text{LF}_{5,-2}[\tilde{\mu}] + \frac{2}{3} g_2^4 \text{LF}_{2,1,0}[\mathbf{m}_2, \tilde{\mu}] + \\
& \frac{17}{12} g_2^4 \text{LF}_{2,2,-1}[\mathbf{m}_2, \tilde{\mu}] - \frac{1}{3} g_2^4 \text{LF}_{3,1,-1}[\mathbf{m}_2, \tilde{\mu}] - \frac{7}{3} g_2^4 \text{LF}_{3,2,-2}[\mathbf{m}_2, \tilde{\mu}] + \\
& \frac{3}{2} g_2^4 \mathbf{m}_2^2 \text{LF}_{3,2,-1}[\mathbf{m}_2, \tilde{\mu}] + g_2^4 \text{LF}_{3,3,-3}[\mathbf{m}_2, \tilde{\mu}] - g_2^4 \mathbf{m}_2^2 \text{LF}_{3,3,-2}[\mathbf{m}_2, \tilde{\mu}] + \\
& g_2^4 \text{LF}_{4,2,-3}[\mathbf{m}_2, \tilde{\mu}] - g_2^4 \mathbf{m}_2^2 \text{LF}_{4,2,-2}[\mathbf{m}_2, \tilde{\mu}] - \frac{1}{4} g_2^2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{r}}] + \\
& \frac{1}{6} g_2^2 (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{5,1,-2}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{r}}] - \frac{3}{4} g_2^2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] + \\
& \frac{1}{2} g_2^2 (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{5,1,-2}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{r}}] + \frac{3}{4} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{2,1,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}] - \\
& \frac{3}{4} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}] - \frac{1}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{2,2,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] + \\
& \frac{3}{4} g_2^2 (-\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,1,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] + \\
& \frac{1}{4} g_2^2 (2 \overline{a_u^{\text{pr}}} a_u^{\text{pr}} - \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] + \\
& \frac{3}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} - \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}] + \frac{3}{4} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{2,1,0}[\mathbf{m}_q^{\text{s}}, \mathbf{m}_q^{\text{p}}] - \\
& \frac{3}{4} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1}[\mathbf{m}_q^{\text{s}}, \mathbf{m}_q^{\text{p}}] + \frac{1}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,1,0}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \\
& \frac{1}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,2,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \\
& \frac{3}{4} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{4,1,-1}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] + \\
& \frac{1}{2} g_2^2 (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{5,1,-2}[\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}] - \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{2,1,0}[\mathbf{m}_u^{\text{t}}, \mathbf{m}_d^{\text{r}}] + \\
& \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1}[\mathbf{m}_u^{\text{t}}, \mathbf{m}_d^{\text{r}}] + \frac{5}{12} g_1^2 g_2^2 \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_1] - \frac{1}{3} g_1^2 g_2^2 \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_1] - \\
& \frac{4}{3} g_2^4 \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_2] - \frac{5}{3} g_2^4 \text{LF}_{3,2,-2}[\tilde{\mu}, \mathbf{m}_2] + \frac{9}{4} g_2^4 \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_2] + \frac{1}{2} g_2^4 \text{LF}_{4,2,-3}[\tilde{\mu}, \mathbf{m}_2] - \\
& \frac{1}{2} g_2^4 \tilde{\mu}^2 \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_2] - g_2^4 \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_2] + 2 g_1^2 g_2^2 \text{LF}_{2,2,1,-2}[\mathbf{m}_2, \tilde{\mu}, \mathbf{m}_1] + \\
& \frac{3}{2} \mathbf{m}_1 \mathbf{m}_2 g_1^2 g_2^2 \text{LF}_{2,2,1,-1}[\mathbf{m}_2, \tilde{\mu}, \mathbf{m}_1] - g_1^2 g_2^2 \text{LF}_{3,2,1,-3}[\mathbf{m}_2, \tilde{\mu}, \mathbf{m}_1] - \\
& \mathbf{m}_1 \mathbf{m}_2 g_1^2 g_2^2 \text{LF}_{3,2,1,-2}[\mathbf{m}_2, \tilde{\mu}, \mathbf{m}_1] - \frac{1}{4} \tilde{\mu}^2 \overline{y_e^{\text{pr}}} \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \text{LF}_{2,2,1,-1}[\mathbf{m}_l^{\text{p}}, \mathbf{m}_l^{\text{s}}, \mathbf{m}_e^{\text{r}}] + \\
& \frac{3}{4} (\overline{y_u^{\text{st}}} y_u^{\text{pt}} \overline{a_d^{\text{pr}}} a_d^{\text{sr}} - \tilde{\mu}^2 \overline{y_d^{\text{pr}}} \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}}) \text{LF}_{2,2,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_d^{\text{r}}] + \\
& \frac{3}{4} (\overline{y_d^{\text{st}}} y_d^{\text{pt}} \overline{a_u^{\text{pr}}} a_u^{\text{sr}} - \tilde{\mu}^2 \overline{y_u^{\text{pr}}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}}) \text{LF}_{2,1,1,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_u^{\text{r}}] + \\
& \frac{3}{4} (-\overline{y_d^{\text{st}}} y_d^{\text{pt}} \overline{a_u^{\text{pr}}} a_u^{\text{sr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}}) \text{LF}_{3,1,1,-1}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_u^{\text{r}}] - \\
& \frac{3}{4} \tilde{\mu}^2 \overline{y_u^{\text{pr}}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \text{LF}_{2,1,1,0}[\mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_u^{\text{t}}] + \frac{3}{4} \tilde{\mu}^2 \overline{y_u^{\text{pr}}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \\
& \text{LF}_{3,1,1,-1}[\mathbf{m}_q$$